

Prüfung Autonomes Fliegen

Aufgabe 1

Simulation

The complete nonlinear 6-DOF (six degrees of freedom) model of the quadrotor is described by

$$\ddot{x} = (\cos(\phi) \sin(\theta) \cos(\psi) + \sin(\phi) \sin(\psi)) \frac{U_1}{m} \quad (1)$$

$$\ddot{y} = (\cos(\phi) \sin(\theta) \sin(\psi) - \sin(\phi) \cos(\psi)) \frac{U_1}{m} \quad (2)$$

$$\ddot{z} = -g + (\cos(\phi) \cos(\theta)) \frac{U_1}{m} \quad (3)$$

$$\ddot{\phi} = \frac{\ell}{I_x} U_2 \quad (4)$$

$$\ddot{\theta} = \frac{\ell}{I_y} U_3 \quad (5)$$

$$\ddot{\psi} = \frac{1}{I_z} U_4 \quad (6)$$

in which $[x, y, z]^T$ denotes the position of center of mass represented in the inertial coordinate system O_I , m is the mass of the quadrotor, g denotes the magnitude of north gravity. Furthermore, $[\phi, \theta, \psi]^T$ represents the orientation.

Aufgabe 1-1

Write the nonlinear model in state-space form as:

$$\dot{X} = f(X, U)$$

where $X = [x_1, x_2, \dots, x_{12}]^T \in \mathbb{R}^{12}$ is the state vector defined as:

Term	Description
x, y, z	Position of the quadrotor's center of mass in O_I .
m	Mass of the quadrotor.
g	Magnitude of the gravitational acceleration.
ϕ, θ, ψ	Orientation of the quadrotor (roll, pitch, and yaw angles).
U_1	Total thrust generated by the quadrotor.
U_2	Rolling moment control input.
U_3	Pitching moment control input.
U_4	Yawing moment control input.
I_x, I_y, I_z	Moments of inertia about the x , y , and z axes, respectively.
ℓ	Distance from the center of mass to each rotor axis (arm length).

Table 1: Description of terms in the nonlinear 6-DOF quadrotor model.

$$\begin{aligned}
x_1 &= \phi, & x_7 &= z, \\
x_2 &= \dot{\phi}, & x_8 &= \dot{z}, \\
x_3 &= \theta, & x_9 &= \dot{\theta}, \\
x_4 &= \dot{\theta}, & x_{10} &= \dot{\phi}, \\
x_5 &= \psi, & x_{11} &= \dot{\psi}, \\
x_6 &= \dot{\psi}, & x_{12} &= \dot{\theta}.
\end{aligned}$$

The inputs to the system are represented by:

$$U = [U_1, U_2, U_3, U_4]^T,$$

where U_1 is the thrust, and U_2, U_3, U_4 are the control torques for roll, pitch, and yaw, respectively.

Aufgabe 1-2

Write the altitude dynamics in the form:

$$\dot{x} = Ax + B(x)U_1.$$

Then, linearize the nonlinear subsystem around the hover condition:

$$x_1 = x_2 = x_3 = x_4 = x_5 = x_6 = x_8 = x_9 = x_{10} = x_{11} = x_{12} = 0,$$

with the z -coordinate $x_7 = 1$ meter.

Aufgabe 1-3

Assuming a mass of 1 kilogram and a gravitational acceleration of 9.81 m/s^2 , analyze the controllability of the linearized altitude system.

Aufgabe 2

Linear and Nonlinear Control

Consider the altitude dynamics of the 6-DOF system described in Question 1.

$$\ddot{z} = -g + (\cos(\phi) \cos(\theta)) \frac{U_1}{m}$$

Aufgabe 2-1

Design a PD controller for the control input U_1 to regulate the altitude dynamics.

Aufgabe 2-2

Prove that the following control law, designed using Backstepping control and an appropriately chosen Lyapunov function, ensures that the altitude dynamics are globally asymptotically stable.

$$U_1 = \frac{m}{\cos(x_1) \cos(x_3)} (z_1 + g - \alpha_1(z_2 + \alpha_1 z_1) - \alpha_2 z_2), \quad (7)$$

where α_1 and α_2 are positive constants, we have:

$$z_1 = x_{7_d} - x_7 \quad (8)$$

$$z_2 = x_8 - \alpha_1 z_1 \quad (9)$$

Aufgabe 3

Path Planning

A quadrotor must move along a straight line from an initial position $q_i = 0 \text{ m}$ to a final position $q_f = 10 \text{ m}$. The system has the following constraints:

- Maximum acceleration: $q_{\max} = 2 \text{ m/s}^2$
- Maximum velocity: $q_{\max} = 3 \text{ m/s}$

Aufgabe 3-1

Calculate the time required for each phase:

- Acceleration time (t_a): The time to reach $q_{\max}$ from rest.
- Deceleration time (t_d): The time to decelerate from $q_{\max}$ to rest.
- Total time (T) and the duration of the constant velocity phase.

Aufgabe 3-2

Derive the position equations for each phase:

- Acceleration phase: $q(t)$ for $t \in [0, t_a]$.
- Constant velocity phase: $q(t)$ for $t \in [t_a, T - t_a]$.
- Deceleration phase: $q(t)$ for $t \in [T - t_a, T]$.

Aufgabe 3-3

Calculate and verify the total displacement to ensure it matches $q_f = 10 \text{ m}$.

Aufgabe 3-4

Sketch the trajectory:

- Plot $q(t)$ and $\dot{q}(t)$ vs. t for the entire trajectory, labeling key points ($t_a, T - t_a, T, q_{\max}$, etc.).

Aufgabe 4

Design a 2D trajectory for a quadrotor to move from Point $A(0, 0)$ to Point $B(10, 10)$ in a total time $T = 10$ seconds. The trajectory should minimize the **jerk** (the derivative of acceleration) with the following boundary conditions:

- Initial position $\mathbf{r}(0) = (0, 0)$ and final position $\mathbf{r}(T) = (10, 10)$.
- Initial and final velocities $\mathbf{v}(0) = \mathbf{v}(T) = (0, 0)$.
- Initial and final accelerations $\mathbf{a}(0) = \mathbf{a}(T) = (0, 0)$.

Aufgabe 4-1

For the functional, use a polynomial trajectory representation for the $x(t)$ and $y(t)$ coordinates of the quadrotor.

Aufgabe 4-2

Derive the polynomial coefficients that minimize jerk.

Aufgabe 4-3

Calculate the resulting trajectories $x(t)$ and $y(t)$, along with their first and second derivatives $(\dot{x}(t), \dot{y}(t))$ and $(\ddot{x}(t), \ddot{y}(t))$. Plot the following:

1. The trajectories $x(t)$ and $y(t)$ as functions of time.
2. The first derivatives $(\dot{x}(t), \dot{y}(t))$ over time.
3. The second derivatives $(\ddot{x}(t), \ddot{y}(t))$ over time.
4. The 2D path of the quadrotor from Point $A(0,0)$ to Point $B(10,10)$.

Aufgabe 5

Navigation

Design both an open-loop and a closed-loop observer tailored to the altitude dynamics described in Question 1.